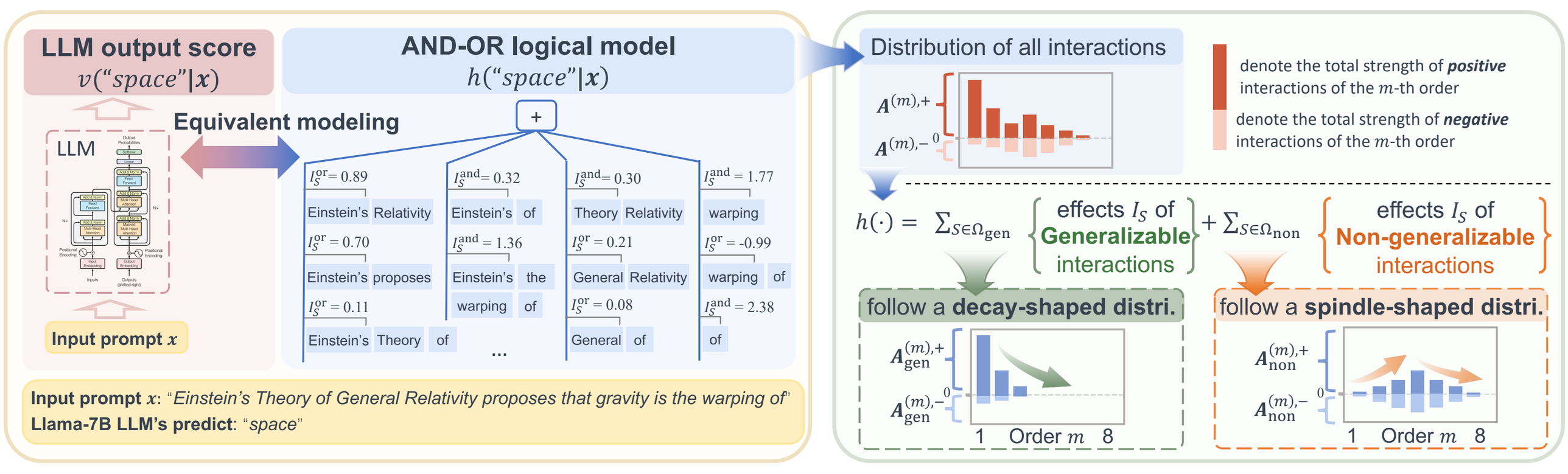

LLM output score
$v(\text{“space”}|\boldsymbol{x})$
LLM
Input prompt $\boldsymbol{x}$
Equivalent modeling
AND-OR logical model
$h(\text{“space”}|\boldsymbol{x})$
+
$I_S^{\text{or}} = 0.89$
Einstein's Relativity
$I_S^{\text{or}} = 0.70$
Einstein's proposes
$I_S^{\text{or}} = 0.11$
Einstein's Theory of
$I_S^{\text{and}} = 0.32$
Einstein's of
$I_S^{\text{and}} = 1.36$
Einstein's the
warping of
...
$I_S^{\text{and}} = 0.30$
Theory Relativity
$I_S^{\text{or}} = 0.21$
General Relativity
$I_S^{\text{or}} = 0.08$
General of
$I_S^{\text{and}} = 1.77$
warping
$I_S^{\text{or}} = -0.99$
warping of
$I_S^{\text{and}} = 2.38$
of
Input prompt $\boldsymbol{x}$: “Einstein's Theory of General Relativity proposes that gravity is the warping of”
Llama-7B LLM's predict: “space”
Distribution of all interactions
$A^{(m),+}$
$A^{(m),-}$
0
denote the total strength of positive interactions of the m-th order
denote the total strength of negative interactions of the m-th order
$h(\cdot) = \sum_{S\in\Omega_{\text{gen}}}$ effects I_S of Generalizable interactions $+ \sum_{S\in\Omega_{\text{non}}}$ effects I_S of Non-generalizable interactions
follow a decay-shaped distri.
$A_{\text{gen}}^{(m),+}$
$A_{\text{gen}}^{(m),-}$
0
1 Order m 8
follow a spindle-shaped distri.
$A_{\text{non}}^{(m),+}$
$A_{\text{non}}^{(m),-}$
0
1 Order m 8